Silicon Sampling on CES 2025: Where We Stand

Two branches, one benchmark · prompting vs fine-tuning · what holds up

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In brief. Both branches try to reproduce how 15 Canadian subgroups (5 regions × 3 age groups) answered 10 CES 2025 questions, using only data from before 2025. Fine-tuning works much better than prompting (18 vs 30 answers out of 100 in the wrong place). But **a four-year-old poll, CES 2021 reused unchanged, still beats both (15 out of 100)**. Both branches find the same reason: models get the direction of subgroup differences right but shrink them to about a third of their real size.

1. The two branches

Same test for both: 10 CES 2025 questions × 15 subgroups = 150 answer distributions, each compared with the real CES 2025 one (N = 20,180). **Score: total variation (TV)**, the share of answers that would have to move to match reality; 0.18 = 18 out of 100 in the wrong place. Lower is better.

	Justin: prompting	**Hubert: fine-tuning**
Model	Qwen 3.5 9B, as is (Together API)	Llama 3.3 70B, LoRA (Azure AI Foundry)
Training	None	50k answers from 8 surveys, 2019–2024
Synthetic respondents	CES 2021 profiles, 100 per subgroup	Democracy Checkup 2024 profiles, 1,000 per subgroup
What the model sees	Demographics + party ID, left–right, 2019 vote	The person's own answers to 7 fixed questions (ideology, party, trust…)
Calls	1,500 (10 questions per call)	150,000 (1 question per call)

Dropped along the way (tested, not worth keeping):

- *Bigger model* (Qwen 2.4T): worse than the 9B in every condition; it piles onto neutral answers.
- *Thinner profiles* (demographics only, or no party information): party information is the largest single gain for prompting.
- *Semantic context* (picking the past answers closest in meaning): worse than fixed themes at every size.
- *More training data* (100k): no better than 50k.
- *IRT slope correction* (stretching model gaps ×3): gains 0.01 overall and hurts 4 of 10 questions. Even fed the real 2025 national figures, it does worse (0.098) than CES 2021 gaps (0.064).

2. Scoreboard

Mean TV over the 150 subgroups, and over the 10 national distributions. Green is closer to reality, red further. Grey rows use CES 2025 itself and only show what is reachable.

Source	Branch	By subgroup	National
Human floor (two halves of CES 2025)	Reference	0.062	0.062
Real 2025 national for every subgroup	Reference	0.085	0
CES 2021, same subgroup, unchanged	Past poll	0.151	0.141
CES 2021 multinomial logit (rich + partisan)	Past poll + model	0.175	0.150
Fine-tuned Llama 3.3 70B (fixed context, 50k)	Hubert	0.178	0.157
Prompted Qwen 3.5 9B (rich + partisan)	Justin	0.302	0.258
Untrained Llama 3.3 70B	Hubert	0.325	0.296

- **Fine-tuning roughly halves the error** of an untrained model (0.325 → 0.178) and does much better than the best prompting (0.302).
- **A fine-tuned model is about as good as a logit trained on 2021** (0.178 vs 0.175 by subgroup), but neither beats simply reusing CES 2021.

3. By question

Mean TV over the 15 subgroups. **Bold** marks the best of the three forecasts on each question.

Question	CES 2021 unchanged	Fine-tuned Llama 70B	Prompted Qwen 9B	Floor
Spending: affordable housing	**0.08**	0.10	0.13	0.05
Satisfaction with democracy	**0.06**	0.12	0.23	0.05
Medical assistance in dying	**0.09**	0.14	0.23	0.09
Jobs before environment	**0.13**	0.15	0.31	0.10
Spending: defence	0.31	**0.17**	0.24	0.04
Pipelines	**0.16**	0.17	0.31	0.10
Spending: national childcare	**0.09**	0.18	0.31	0.06
Personal finances, past year	**0.13**	0.21	0.21	0.04
Spending: reconciliation	**0.17**	0.24	0.44	0.05
Immigration levels	0.31	**0.30**	0.60	0.04
Mean	**0.151**	0.178	0.302	0.062

- **CES 2021 wins on 8 of 10 questions.** The fine-tuned model wins on **defence**, where opinion moved sharply after 2021 and its training covers surveys up to 2024. On **immigration** both are poor.
- **The fine-tuned model beats prompting on every question**, by the widest margin on immigration (0.30 vs 0.60) and reconciliation (0.24 vs 0.44).
- **Shared misses.** Immigration: 62% of Canadians want fewer immigrants; prompted models chose it 2–4% of the time, and Azure's content filter removed most immigration items from fine-tuning. Reconciliation: prompted models say "spend more" 68–91% of the time, against 22% of Canadians.

4. What both branches find

- **Subgroups are flattened.** Hubert measures model subgroup gaps at about 31% of their real size (correlation 0.42; CES 2021: 69%, 0.71). Justin finds the same with an item response model: right direction on every question, slope at roughly a third of the human one.
- **Most of the error is the national level.** For fine-tuning and CES 2021 alike, the national error (0.157, 0.141) is close to the subgroup error. Giving CES 2021 gaps the real 2025 national figure drops the error to 0.064, almost the human floor (0.062).

- **On repeated questions, a past poll is the bar, and it is high.** All 10 questions were asked in CES 2021. Silicon sampling should add value on questions with no past version.

5. Next steps

1. **Report one score.** Mean TV by subgroup, for every run and baseline.
2. **Test the realistic hybrid:** national level from the fine-tuned model + subgroup gaps from CES 2021. This is the forecast behind the 0.064 ceiling, and it needs no new runs.
3. **Build a novel-question benchmark:** CES 2025 items never asked in 2019 or 2021, scored separately.
4. **Fine-tune without Azure's filter** (another host) to recover immigration.
5. **Repeat the best recipe with a new seed** before treating 50k vs 100k as settled.

Comparability: Justin's subgroup scores use 100 synthetic respondents per subgroup (Hubert: 1,000), which adds a few points of sampling noise; his scores are weighted by population (Hubert: equal weights; 0.174 vs 0.178 for the fine-tuned model). Logit and prompting figures as reported in Justin's notes of October 12–14.